

SPRINGFIELD COMANCHE, CO, USA

WMO: 724646

Lat: 37.283N

Lon: 102.614W

Elev: 4383

StdP: 12.51

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 03028

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.4	9.4	-4.4	5.1	24.6	-0.3	6.4	27.9	32.8	39.6	28.6	37.8	7.9	270	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.3	98.0	63.8	95.4	63.6	92.9	63.5	68.8	84.8	67.9	84.6	66.8	83.6	14.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	107.6	71.6	63.5	103.5	70.5	62.6	100.1	70.4	35.6	84.0	34.7	84.3	33.8	83.7	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.8	27.3	24.6	DB	-6.1	102.4	5.0	1.9	-9.6	103.8	-12.5	104.9	-15.3	106.0	-18.9	107.4	(4)
(5)			WB	-6.7	71.0	4.9	1.3	-10.2	71.9	-13.1	72.7	-15.8	73.5	-19.3	74.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	33.6	34.8	44.7	52.0	61.3	72.6	77.1	74.6	67.5	54.3	42.9	33.5	
	DBStd	18.00	10.21	10.97	10.11	9.10	8.77	6.82	4.98	5.27	7.27	9.44	9.96	10.57	
	HDD50	2098	512	432	223	84	18	0	0	0	1	64	252	512	
	HDD65	5074	974	844	631	395	172	15	1	3	58	344	662	975	
	CDD50	3626	3	8	57	144	367	678	841	763	525	198	40	2	
	CDD65	1128	0	0	0	5	57	242	377	301	132	14	0	0	
	CDH74	15346	1	9	87	322	1173	3312	4723	3466	1790	428	35	0	
	CDH80	7590	0	0	8	72	443	1736	2645	1778	785	121	2	0	
Wind	WSAvg	11.7	10.4	11.5	12.5	14.1	13.1	12.8	11.3	10.6	11.7	11.2	11.1	10.7	
Precipitation	PrecAvg	15.6	0.5	0.4	0.9	1.1	2.2	2.1	2.8	2.6	0.9	0.9	0.5	0.4	
	PrecMax	21.7	1.8	1.3	3.4	3.7	4.6	7.0	8.3	8.3	2.9	3.5	2.4	1.8	
	PrecMin	8.8	0.0	0.0	0.0	0.2	0.4	0.4	0.3	0.5	0.0	0.0	0.0	0.0	
	PrecStd	2.8	0.4	0.4	0.8	0.8	1.2	1.4	1.8	1.8	0.9	1.1	0.6	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.2	74.6	80.9	86.6	92.8	101.2	100.4	99.4	95.5	88.2	78.5	68.9	
		MCWB	45.8	47.4	51.2	52.7	57.8	61.6	65.1	63.7	61.5	56.9	50.3	45.7	
	2%	DB	64.1	67.2	76.4	81.7	89.0	97.2	98.3	96.4	91.6	83.4	73.5	63.2	
		MCWB	43.6	44.5	49.0	51.9	57.3	61.6	64.9	64.5	61.3	55.1	48.7	43.3	
	5%	DB	57.6	61.4	71.4	77.9	85.0	93.6	95.8	93.0	88.6	79.2	68.1	57.5	
		MCWB	40.9	42.2	47.6	51.1	56.3	61.8	64.7	64.4	61.5	54.8	46.4	40.8	
	10%	DB	52.1	54.9	65.4	72.7	81.1	90.5	93.1	90.2	84.6	73.5	61.8	51.9	
		MCWB	38.4	39.6	45.2	49.9	56.3	62.0	64.9	64.6	61.0	53.1	44.3	38.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	48.6	54.2	58.3	64.8	68.8	70.2	70.4	67.9	63.4	52.6	47.4
			MCDB	66.9	68.7	72.3	72.2	79.1	84.6	85.9	84.8	83.8	72.7	70.2	66.2
2%		WB	44.1	45.5	51.1	55.7	62.3	67.1	68.8	68.8	66.0	59.9	50.3	44.1	
		MCDB	61.6	65.2	70.4	70.9	77.7	84.7	85.5	84.0	80.8	73.8	68.3	61.1	
5%		WB	41.5	42.8	48.8	53.4	60.4	65.8	68.0	67.8	64.4	57.0	47.8	41.6	
		MCDB	57.0	60.5	67.4	69.0	76.5	82.8	85.6	83.2	79.0	71.1	66.2	56.3	
10%		WB	38.7	40.1	46.5	51.4	58.6	64.5	67.0	66.7	63.0	54.7	45.2	38.8	
		MCDB	51.5	54.5	64.3	69.4	74.3	81.6	85.0	82.0	77.7	71.0	61.8	51.2	
Mean Daily Temperature Range		5% DB	MDBR	28.2	28.2	30.4	29.4	29.4	29.2	28.3	26.8	28.0	29.3	29.2	26.6
			MCDBR	36.6	38.8	38.1	38.2	35.1	33.8	32.5	31.9	33.4	36.4	37.4	35.5
	MCWBR		20.1	20.1	17.4	16.2	13.3	10.8	8.8	9.0	11.1	15.3	17.8	19.6	
	5% WB	MCDBR	35.0	37.0	35.3	31.8	30.6	28.9	27.6	26.6	27.2	29.5	34.4	33.5	
		MCWBR	19.7	19.6	17.0	15.0	13.2	11.0	8.6	9.1	10.1	14.7	17.2	19.2	
Clear-Sky Solar Irradiance	taub	0.245	0.253	0.278	0.305	0.323	0.346	0.363	0.347	0.318	0.283	0.258	0.246		
	taud	2.593	2.588	2.515	2.447	2.457	2.421	2.427	2.440	2.503	2.579	2.573	2.585		
	Ebn at Noon	309	318	316	310	303	295	289	292	297	301	300	301		
	Edh at Noon	23	26	31	35	36	37	36	35	31	26	23	21		
All-Sky Solar Radiation	RadAvg	908	1171	1598	1963	2264	2428	2377	2105	1778	1341	998	793		
	RadStd	68	102	123	107	124	102	85	92	87	115	72	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		-0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	N/A

Nomenclature: See separate page